

Profits From Technical Trading Rules

February 5, 1997

Mark J. Ready
University of Wisconsin-Madison
975 University Avenue
Madison, WI 53706
608-262-5226
mjready@facstaff.wisc.edu

Abstract

I examine technical trading rules developed by Allen and Karjalainen (1993) and moving average rules studied by Brock, Lakonishok, and LeBaron (1992). Using intraday data, I find that profits reported in these studies may not have been achievable due to price slippage between when trading signals are measured and when trades could actually be executed. I also find that the 0.1% one-way transaction cost assumed by Allen and Karjalainen appears too low. I find that the ability of all of the rules to distinguish between days of high and low returns has declined dramatically over the last six to eight years, suggesting that factors causing return predictability may change over time.

I test modified rules that account for higher transaction costs and potential price slippage and are based on more recent data. These new rules do considerably better, but they still are unable to beat a buy and hold strategy.

Can technical trading rules really make money? In “Using Genetic Algorithms to Find Technical Trading Rules” by Allen and Karjalainen (1993) (hereafter AK), the authors develop technical trading rules that switch between stocks and T-Bills. They estimate that these rules would have significantly outperformed a stock buy-and-hold strategy, even after transaction costs. In calculating their profit estimates, however, AK ignore dividends and assume that each simulated trade could be made at the previous day’s close, plus or minus a fixed 0.1% transaction cost. These simulated trade price estimates could be inaccurate if the quotes have moved since the last trades were recorded for at least some of the securities in the portfolio, which could be particularly problematic for momentum strategies. Also, estimating transaction costs using transaction prices and a fixed adjustment may be inaccurate if spreads tend to widen at the same time the trading decisions are made, which would be true if spreads respond to the same factors that are being used to generate the trading decisions.

I begin by reevaluating the profitability of the two representative rules that are explicitly defined by AK, as well as the profitability of three long-term moving average rules described in and Brock, Lakonishok and LeBaron (1992) (hereafter BLL). AK find their rules generate annual excess returns of more than 4% compared to a buy-and-hold strategy over the 1970-1989 period. AK also find that the BLL moving average rules generate annual excess returns of about 2-3% over this same period. I use data that include dividends and I show that there is little relation between the timing of dividend payments and the trading position of the rules. The inclusion of dividends does change the excess returns, however, because whenever the technical trading rule portfolios are invested in T-Bills, they miss out on the dividends earned by the stock buy-and-hold portfolio. I show that the inclusion of dividends reduces all of the annual excess return estimates by about 2%. I also address the possibility of price slippage by calculating the profitability of the rules assuming trades are made at the closing price on the day following the buy signal. I show that this change completely eliminates the excess return from the AK rules, not only for the full 1970-1989 sample, but in every 5-year subperiod. The change also eliminates the excess returns for two of the three BLL moving average rules for the 1970-1989 sample. The other remains marginally profitable for the full sample, but all of these profits stem from the 1970-1974 subperiod.

In order to assess the accuracy the 0.1% transaction cost estimate used by AK, and to see how quickly the price-slippage occurs, I use intraday trade and quote data from 1988-1992. I use the methodology developed in “Estimating the Profits from Trading Strategies,” by Knez and Ready (1996), as well as some additional results from “The

Specialist's Discretion: Stopped Orders and Price Improvement," by Ready (1996) to estimate execution prices conditioning on quoted spread, depth and most recent transaction price. Our results are based on intraday trade and quote data for the years 1988-1992, and reflect both changes in the quotes since the previous close and transaction costs resulting from the bid-ask spread.

Even if trades could have been made at the previous days close, all of the rules were unprofitable during the 1988-1992 period. Using the next day's close to account for price slippage makes them even worse. This means that the intraday results from this period may not be representative of what could have been achieved earlier in the sample, because whatever factors caused the earlier profitability of these rules may have disappeared before 1988. In any case, I find that the price slippage observed between closing prices in the 1988-1992 sample tends to occur before 11:00 am. Using our conditional execution prices to simulate the profits from the strategies, I find that a portfolio with a \$10,000,000 initial investment (about \$35,000 per stock) generates one-way transaction costs of about 0.12%, marginally greater than the 0.1% assumed by AK. I also estimate that a portfolio of \$100,000,000 (\$350,000) per stock would generate transaction of about 0.3%. AK report that one way transaction costs would have to be less than 0.18% (assuming no price slippage from the previous close) for the average of the excess returns for their 100 best rules to remain positive over the 1970-1989 period.

I propose and test modifications to the AK genetic algorithm procedure, which account for higher transaction costs and potential price slippage, and are based on more recent data. Not surprisingly, I find that modifying the procedure to reflect dividends results in rules that are more frequently invested in stocks, and that modifying the procedure to incorporate higher transaction costs results in rules that make fewer trades. If it weren't for price slippage, it appears that the appropriately modified rules could still have been profitable during the 1970-1989 period. After recognizing price slippage, however, it appears that even the rules that were "trained" to take this problem into account are unable to outperform the buy and hold strategy.

The modified rules do seem to do better in the earlier years of the 1970-1989 test period, suggesting that the patterns that were "learned" from late 1960's during the rule development may have slowly disappeared. If the patterns were replaced by new ones, then rules "trained" on the most recent data should tend to do better. I test this by generating another set of rules using the 1980's for training and selection. Using 1990-1995 as a test period, I find that the rules developed based on the 1980's outperform those that were developed based on the 1960's. However, neither set of rules is able to outperform the simple buy and hold strategy.

The remainder of the paper is organized as follows. Section 1 describes the trading rules tested by BLL and AK and describes the AK procedure for generating rules. In Section 2, I describe the data and estimation procedures. Section 3 tests the AK and BLL rules on daily data, and Section 4 uses intradaily data to get a better estimate of transaction costs and price slippage. Section 5 considers rules based on various modifications to the AK procedure, and Section 6 concludes.

1. The Trading Rules:

The trading rules examined by BLL are based on the idea that the market develops “support” and “resistance” levels, and once the price “breaks through” these levels it is likely to continue in the same direction for a while. Their trading rules use simple comparisons of a short run moving average (in many cases, just the most recent price) to a longer run moving average (from 50 to 500 trading days).¹ A buy signal is triggered whenever the short run moving average is above the long run moving average by a pre-specified percentage, and a sell signal is triggered when the reverse occurs. After a sell signal is triggered, the portfolio’s assets are moved from stock into risk-free bonds. When the next buy signal is triggered, the assets are moved from bonds into stocks.

In this paper, I use the moving average rules denoted (1,50,1), (1,150,1) and (1,200,1) by both BLL and AK. The first number is the number of days used in the short moving average, so these rules use the most recent closing price. The second number gives the number of days in the long moving average, and the third number is the percentage that the price must differ from the long moving average in order to generate a signal.

The rules developed by AK also use moving averages as building blocks, but the buy and sell signals can depend in complex ways on the past sequence of observed prices. Their genetic algorithm procedure constructs trading rules by assembling simple Boolean and real-valued functions in a tree structure.² Initially, 500 rules are randomly generated and tried out on the 1964-1967 S&P price data, which they refer to as the “training period.” The fitness of each rule is measured as the excess return in this training period, and these excess return calculations include a .1% transaction cost whenever the rule makes a trade. A new “generation” of 500 rules is then constructed, wherein each rule in the new generation results from a random choice of two “parents” from the initial population with the selection probabilities skewed toward the fittest rules. Parts of the tree structure of each parent are combined to make each new rule, and the new rule randomly replaces one of the rules in the population, where the replacement choice is skewed toward the rules with lower rankings. The best rule from each generation is tried on the 1968-1969 S&P data,

which is called the “selection period,” and the process of making new generations stops once the improvement in the selection period levels off. The “best” rule is the one with the highest excess return during the selection period.

AK repeat the process of generating a best rule 100 times and then calculate the excess returns for their 100 best rules using the 1970-1989 S&P data, which they call the “test period.” They find that their rules produce annual excess returns of about 4%, as compared to 2-3% available from the rules used by BLL. AK provide explicit definitions for two of their 100 best rules (rules 4 and 7), which achieve test period profitability that is representative of all of their rules. It is these two rules that I examine in this paper.

2. Data and Procedures

I simulate the profits from technical trading rules using a portfolio is comprised of the largest 20% of the stocks on the NYSE, using market capitalization measured as of June 30 of each year. Portfolio weights are also proportional to market capitalizations. This portfolio is highly correlated with the S&P 500 portfolio used by Allen and Karjalainen (1993), but I restrict it to NYSE stocks because the conditional execution price relationships were developed using only NYSE stocks. The S&P index data used by AK exclude dividends, so to check our results against those reported by AK, I also calculate returns on our portfolio excluding cash dividends. Moreover, since the rules of both AK and BLL are “designed” to calculate buy and sell signals from an index that excludes dividends, I construct an index of the stock in our portfolio that excludes dividends in order to calculate our buy and sell periods for each trading rule. I then use the data including dividends to evaluate the results from using each rule.

Closing prices, shares outstanding, and dividend information for 1970-1995 are from the Center For Research in Securities Prices (CRSP), and intraday trade and quote information for 1988-1992 are from the Institute for the Study of Securities Markets (ISSM). Risk-free rate data are based on the annualized one month T-Bill yields from the CRSP bond files. Price improvement relationships are based on the Trades, Orders, Reports, and Quotes (TORQ) database available from the New York Stock Exchange. The TORQ database covers a sample of 144 NYSE firms for the 3-month period from November 1990 to January 1991.

In each of the trading rules, closing prices generate buy or sell signals. For the 1988-1992 period, I estimate the expected execution prices using trade and quote information measured as of 11:00 on the following day. By using the current quote, I capture the latest market information. This allows for the possibility that the market participants are responding in some way to the same price pattern that caused the trading

rule to issue a buy or sell signal. Of course, it is well known that for NYSE stocks that the current bid or ask price is generally a conservative estimate of the expected execution price, because many orders actually do better than the quote that was in effect when the order was submitted. This difference between the quote and the execution price is called “price improvement” and Knez and Ready (1996) show that expected price improvement is significantly and non-linearly related to “excess depth” which they define as the difference between the quoted depth and the size of the order, divided by the average quoted depth for the firm.

Knez and Ready (1996) estimate the functional dependence of expected price improvement on excess depth using a nonlinear regression technique called loess. Briefly, the loess technique estimates the expected value of the dependent variable at discrete values of the independent variable, called “grid points.” At each grid point, the estimate is the predicted value at the grid point from a weighted least squares regression with highest weights given to the observations that are closest to the grid point. These weights are determined by a spanning parameter, which is simply the fraction of the total sample used in calculating the value at each grid point, and a kernel function. The kernel function causes the weights to decline from one, for observations extremely close to the grid point, to zero for the observation that is the farthest away from the grid point of those that are used in the estimate (as determined by the spanning parameter).

Ready (1997) extends the analysis of Knez and Ready (1996) to show that the position of the last trade price relative to the current quote provides additional information in addition to the excess depth for estimating expected price improvement. This position is defined as “same side” if the last trade was at or below (above) the current bid (ask) and the incoming market order is a sell (buy). “Opposite side” is defined analogously, and the “midpoint” position occurs whenever the last trade was at the midpoint of a current $\$1/4$ spread, regardless of the direction of the incoming order. Figure 1 below shows the loess estimates of expected price improvement as a function of excess depth for all market orders for the largest 20% of the firms in the TORQ sample that arrived at the NYSE via the SuperDOT system when the spread between the quoted bid and ask prices was $\$1/8$ (left panel) or $\$1/4$ (right panel). A separate smooth is plotted for each last trade position.

The curves in Figure 1 are plotted over excess depths ranging from $-.5$ to 2.0 , primarily to focus on the most interesting feature of the relationship, but this range does capture about 85% of the market orders in the sample with positive excess depth and about 95% of the orders with negative excess depth. As shown by Knez and Ready (1996), expected price improvement tends to be fairly constant in the region of positive excess depths, but declines dramatically as excess depth approaches zero (as the order size

approaches the depth). As shown by Ready (1997), expected price improvement is lowest when the last trade was on the “same side” as the current quote.

In simulating execution prices for the trading strategies, I use the smoothes estimated over the entire range of excess depths. In addition to the five smoothes depicted in figure 1, I estimate four smoothes for \$3/8 spreads based on the position of the last trade price. I use the same “same side” and “opposite side” definitions as above, and I define the last trade position as “same inside” if the last trade was \$1/8 above (below) the current bid (ask) and the current order is a sell (buy). The definition of last trade position for the fourth smooth, “opposite inside,” is defined analogously. To determine execution prices for the simulated trades made by our strategies, I measure the spread, excess depth, and last trade position as of 11:00 using the ISSM data. Each simulated purchase (sale) trade’s execution price is equal to the ask (bid) less (plus) the expected price improvement given the excess depth, taken from the appropriate smooth for given the spread and last trade position.

This methodology effectively controls for the market’s dynamic response to some or all of the same information that may be driving a particular trading strategy. The one missing component that can not be handled is the market’s response to the trades themselves. That is, if one had really followed the trading strategy, then one’s own trading activity might impact future market prices. Any simulation using historical data, including the previous studies, will also suffer from this same weakness.

3. Results Using Daily Data

Table 1 shows the results for the 5 technical trading rules using the same methodology as AK over the same time period. The cumulative log returns for each strategy are calculated using (see AK equation 2):

$$r = \sum_{i=1}^T r_t I_b(t) + \sum_{i=1}^T f_t I_s(t) + n \log \frac{1-c}{e^{1+c}}$$

where:

T = number of trading days in the evaluation period

r_t = log stock portfolio return on trading day t

f_t = log T-Bill yield for trading day t (converted to a daily rate)

$I_b(t)$ = 1 if the rule is invested in stock on trading day t, 0 if in T-Bills

$I_s(t)$ = 1 if the rule is invested in T-Bills on trading day t, 0 if in stock

n = number of round-trip transactions made by the rule over the entire period

c = one-way transaction cost (= .001)

Annualized log returns are calculated by dividing the cumulative log return by the number of years in the sample period. Excess returns are calculated by subtracting the annualized log return of the buy-and-hold stock portfolio from the annualized return for the trading rule. The first line in Panel A of Table 1 shows the results for the 5 trading rules over the 1970-1989 period using S&P data, as reported by AK. The second line shows the results for the same rules over the same period using an index constructed from the top 20% of NYSE stocks excluding dividends. The results from the first two lines are quite similar, as would be expected given the large overlap between the S&P stocks and the large NYSE stocks.

The third line of the table shows the results using the same trading decisions, but including dividends in the return calculations. Including dividends lowers the annual excess returns for the 5 rules by about 2%, because the trading rule portfolios don't earn dividends when they are invested in T-Bills. Panel B of Table 1 shows the annualized log dividend yield on stocks averaged over the periods where the strategies are invested in stocks and averaged separately over the periods where the strategies are invested in T-Bills. Dividends do tend to cluster somewhat during the year, but the data in Panel B show that the trading rules do not seem to be avoiding the relatively high dividend days. The strategies are invested in stocks between 47% (AK rule 4) and 65% (moving average rule (1,200,1)) of the days, so they miss out on roughly half of the 4% average annual dividend yield.

Panel C of Table 1 shows the results for 4 subperiods of the 1970-1989 period shown in Panel A. In addition I report the results for the 1990-1995 period, which represent new data beyond those given by AK, and the results for the 1988-1992 period for which I have intraday data. The subperiod data in Panel C of Table 1 indicate that much of the excess returns from the 5 trading rules can be attributed to the first subperiod, when the return on the overall market was slightly negative. This is as might be expected, because a buy and hold T-Bill strategy would have out-performed the market over this period. One would also expect that the strategies would have trouble keeping up with the market in good years, because part of the time they are invested in T-Bills.

Using long-term historical averages, the 1990-1995 period was not an unusual one for the market as a whole. Panel C of Table 1 shows that the 5 technical rules turned in some of their worst performances during this period. This reflects a general decline in the ability of the rules to pick the "right" days to be invested in stocks. Table 2 shows the number of days each strategy was invested in stocks and T-Bills, and the annualized log stock returns averaged over the periods where the strategies are invested in stocks and

averaged separately over the periods where the strategies are invested in T-Bills. These returns are not adjusted for transactions costs.

The first section of Table 2 shows that for the full 1970-1989 period, all of the strategies tended to pick better days to be invested in stocks and to systematically avoid the days with lower stock returns. The differences between the stock returns for days when the strategies were invested in stocks and days when the strategies were invested in T-Bills are much larger for the AK strategies than for the BLL strategies, but at the cost of much more trading. The AK strategies average more than 30 round trips per year, compared with 2-3 for the BLL moving average strategies.

The subperiod results in Table 2 show a decline in the differences between the days when the strategies are invested in stocks and the days that they are invested in T-Bills. In fact, in the most recent 1990-1995 period, the moving average strategies picked the wrong days on average. The AK strategies showed some ability to pick the right days to be invested in stocks during this period, but this ability was clearly much less than shown early in the sample. Interestingly, during the 1988-1992 period for which I have intraday data, on average all 5 of the strategies picked the wrong days to be invested in stocks.

If one were to have actually implemented the strategies over these time periods, it would have been necessary to observe the price signals and then enter orders to execute the trades. The strategies use closing prices, so the earliest one could trade would be at the open on the next morning. There is the possibility that the prices available the next day might have moved since the previous close. If this price movement were systematic, then at least some of the excess returns reported in Table 1 could be illusory. There are two reasons to suspect that this kind of “price slippage” might occur. First, if the closing prices reflect transactions that occurred at different points in time, then this non-synchronous trading would impart autocorrelation to the measured returns on the stock portfolio with no associated trading opportunity. Second, if orders are positively serially correlated, buying (or selling) interest at the close would be followed by more of the same at the following open. The price would adjust at the open in response to the continued imbalance.

As a crude estimate of the potential “price slippage” that may occur between the observation of closing prices and the first real trading opportunity, I use closing prices from the following day. Effectively, this amounts to lagging the trading signals for each rule by one trading day. The results of these calculations are shown in Table 3. These results show that if the following close is a better estimate of the available price than the previous close, then the AK rules would have been unprofitable for all of the subperiods in the sample. By comparing Table 3 to Table 1, it is evident that the effect of this adjustment on the profitability of the AK rules is most pronounced in the 1970-1974 subperiod. The

adjustment has less of an impact on the moving average rules, but it does make the full sample excess returns negative for both the (1,50,1) and (1,200,1) rules. The (1,150,1) rules still shows a small positive excess return during the 1970-1989 period, but it should be noted that 1970-1974 was the only profitable subperiod for this rule.

4. Intraday Results

Table 4 shows the simulated profitability of the 5 trading rules for the 1988-1992 period using the intraday trade and quote data. The results in the first four rows assume trades occur at the previous close (the price used to generate the signal), the most recent trade price as of 11:00, the current quote midpoint as of 11:00 and the closing price as of the end of the day, respectively. All of these calculations assume a fixed .1% one-way transaction cost. The final two rows of the table assume that the trades occur at execution prices estimated from the loess smoothes shown in Section 2. These execution prices will be a function of the order sizes, so it is necessary to assume an initial investment for the portfolios. The \$10 million and \$100 million values used in the table were chosen as potentially interesting levels of investment from the standpoint of a small or medium-sized fund manager.

The returns and excess returns based on the closing prices correspond to the final lines in Panel A of Table 1 and of Table 3, but there are slight differences due to the way the portfolios are formed. Specifically, in order to maximize the use of intraday data, any stock with no good quote as of 11:00 on a day when stocks are to be purchased is excluded from the portfolio.³ Also, the calculations in table 4 include transactions costs on the reinvestment of dividends, whereas those in Tables 1 and 3 assume distributions are costlessly reinvested. Finally, the calculation of the returns using intraday data and current closing prices assume that the first purchase is made on the first day of 1988. The return on this day alone was over 3%, so these portfolios miss out on this high one day return. This explains the slight differences in the buy and hold portfolio returns using the different methods.

Table 4 provides two new results. First, from comparing the excess returns using intraday trade or quote prices with those using closing prices, it is evident that the price slippage that occurs from the previous close until 11:00 is at least as great as the price slippage for the entire day. The second result is that the portfolio returns estimated using the expected execution prices are lower than those estimated using quote midpoints and a fixed one-way transactions cost of 0.1%. This is true even for the smaller investment portfolio. Based on the number of trades made by the various strategies, these results indicate that a one-way transaction cost of 0.12% would be appropriate if the total portfolio

is in the \$10,000,000 range, and closer to .3% if the total portfolio is in the \$100,000,000 range.

5. Extensions

The results above suggest the following modifications to the AK procedure for generating trading rules:

- 1) *Include dividends when evaluating the results.* Otherwise, the selection will be biased toward rules that are more often invested in T-Bills
- 2) *Use a one-way transaction cost of at least 0.12% (consider 0.3%).* The estimate of .1% used by AK appears a little too low, even for a relatively small portfolio investment (by mutual fund standards).
- 3) *Use the next day's closing prices to calculate returns.* This will ensure that the trading profits estimated for the rules will be achievable after accounting for price slippage.
- 4) *Add a function that returns the previous day's portfolio position for the rule.*
The motivation for this extension is discussed below.
- 5) *Update the rules periodically.* The decay of the profitability of the rules in the more recent subperiods seems to indicate that the fundamental market features are evolving over time.

Unlike the others, extension 4 above is not directly motivated by the empirical results. Instead it comes from examination of the rule generation procedure. In principle, this proposed new function could effectively appear in an AK rule because it could be expressed as a combination of their low level functions. In practice, however, the appearance of this construction may be hindered somewhat in the natural selection process of generating new rules, because rules are constrained to have no more than ten levels and no more than 100 nodes. The importance of transactions costs in the ultimate profitability is demonstrated in Section 4. Adding the explicit function to the building blocks for rules could improve the efficiency of the process of generating superior rules.

Table 5 shows the profitability of trading rules that were generated using the same general technique as AK with successive modifications following extensions 1-4 listed above. Every line of the table shows the averages across 100 rules. The standard deviation of each statistic across the 100 rules is reported in parenthesis below each average. The excess returns for each rules are calculated two ways: first using the same assumptions used to generate the rules (the column labeled "rule"), and then using the large firm portfolio including dividends and assuming trades occur at the following day's close

with a one-way transaction cost of .12% (the column labeled “true”). For each rule the average (annualized log) returns are calculated separately for buy and sell days, and a t-statistic is calculated to test whether the difference is greater than zero. The table reports the average (and standard deviation) for these return differences and the average (and standard deviation) of the associated t-statistics.

The first line of Table 5 uses the S&P, and the second uses the large NYSE firm portfolio, both excluding dividends. For these first two lines of the table, the rule generation follows the AK procedure as closely as possible, given the available data⁴ and the results are very close to those reported by AK in their Table 2. The similarity of the first and second lines indicates that the predictability of the large NYSE returns is very similar to the predictability of S&P returns.

The third line of Table 5 shows that including dividends in the stock returns changes the character of the rules (they tend to be invested in stocks more often) without dramatically changing the typical excess returns. There is, however, an increase in the variability of the excess returns across the trials. The fourth line of the table shows that increasing the one-way transaction cost from 0.1% to 0.12% leads to rules that trade somewhat less frequently, but the higher transaction cost still results in lower estimated excess returns. The fifth line of the table shows the results of assuming that trades are executed at the closing price one day after the price used to generate the trading signal. In this case, although the rules do tend to choose higher return days on average, there is no appreciable excess return after transaction costs. Finally, the sixth line of Table 5 shows the results for rules that (may) include a function that returns the current trading position. These rules do tend to trade somewhat less frequently and generate slightly larger returns after transaction costs.

Finally, I consider the potential benefit from periodically generating rules based on more recent information (extension number 5 above). The first five lines of Table 6 show the return data by subperiod for the rules from the final line of Table 5. Like the AK and moving average rules, the profitability of these rules tend to decline over time. These rules were generated using 1963-1967 as the training period and 1968-1969 as the selection period. Their relatively poor performance over the later subperiods indicates that the patterns inherent in the 1960's are no longer present. To address whether there may be new patterns in the more recent periods, the final line of Table 6 shows the returns over 1990-1995 (the final subperiod) for rules that were generated using 1983-1987 as the training period and 1988-1989 as the selection period.

6. Conclusions

Allen and Karjalainen (1993) use a genetic algorithm procedure and stock return data from the late 1960's to develop trading rules. They estimate that these rules would have produced average annual excess returns of 4-5% during the 1970-1989 period, as compared to a buy and hold strategy. In contrast, I estimate that these rules would have produced average annual excess returns of about -10% during this period.

I propose and test modifications to the AK genetic algorithm procedure, which account for higher transaction costs and potential price slippage, and are based on more recent data. Not surprisingly, I find that modifying the procedure to reflect dividends results in rules that are more frequently invested in stocks, and that modifying the procedure to incorporate higher transaction costs results in rules that make fewer trades. If it weren't for price slippage, it appears that the appropriately modified rules could still have been profitable during the 1970-1989 period. After recognizing price slippage, however, it appears that even the rules that were "trained" to take this problem into account are unable to outperform the buy and hold strategy.

The modified rules do seem to do better in the earlier years of the 1970-1989 test period, suggesting that the patterns that were "learned" from late 1960's during the rule development may have slowly disappeared. I test this by generating another set of rules using the 1980's for training and selection. Using 1990-1995 as a test period, I find that the rules developed based on the 1980's outperform those that were developed based on the 1960's. However, neither set of rules is able to outperform the simple buy and hold strategy. It appears that the patterns of the 1960's largely disappeared in later decades, and that there were no new important patterns emerging to take their place.

References:

- Allen, Franklin and Risto Karjalainen, 1993, "Using Genetic Algorithms to Find Technical Trading Rules," working paper 20-93, The University of Pennsylvania.
- Brock, William, Josef Lakonishok and Blake LeBaron, 1992, "Simple Technical Trading Rules and the Stochastic Properties of Stock Returns," *Journal of Finance* 47, 1731-1764.
- Knez, Peter J., and Mark J. Ready, 1996, "Estimating the profits from trading strategies," *Review of Financial Studies* 9, 1121-1164.
- Neely, Christopher, Paul Weber, and Rob Dittmar, 1997, "Is Technical Analysis in the Foreign Exchange Market Profitable? A Genetic Programming Approach" *Journal of Financial and Quantitative Analysis* 32,405-426.
- Ready, Mark J., 1997, "The Specialist's Discretion: Stopped Orders and Price Improvement," working paper, The University of Wisconsin-Madison.

Table 1
Annual log returns and excess returns using daily closing prices

	Buy and Hold	AK rule 4	AK rule 7	Moving Average Rules		
				(1,50,1)	(1,150,1)	(1,200,1)
Panel A: 1970-1989	Return	Excess Returns over Buy and Hold				
Using the S&P Index	.0672	.0453	.0427	.0126	.0310	.0265
-- no dividends						
Using large NYSE --	.0611	.0465	.0387	.0174	.0308	.0319
no dividends						
Using large NYSE --	.1046	.0237	.0197	-.0010	.0136	.0156
including dividends						
Panel B: 1970-1989 Dividend Yields						
When in stocks	.0435	.0437	.0432	.0423	.0418	.0416
When in T-Bills		.0433	.0439	.0451	.0462	.0471
Panel C: Subperiods						
(NYSE incl. dividends)	Return	Excess Returns over Buy and Hold				
1970-1974	-.0244	.1884	.1266	.0866	.1063	.1011
1975-1979	.1280	.0060	.0081	-.0340	-.0424	-.0738
1980-1984	.1300	.0144	-.0327	-.0098	.0029	.0352
1985-1989	.1845	-.1143	-.0225	-.0474	-.0129	-.0005
1990-1995	.1194	-.0996	-.0938	-.0594	-.0236	-.0288
1988-1992	.1465	-.1444	-.1188	-.0780	-.0572	-.0423

Table 2
Comparison of returns during buy and sell periods

	Buy and Hold	AK rule 4	AK rule 7	Moving Average Rules		
				(1,50,1)	(1,150,1)	(1,200,1)
<u>1970-1989</u>						
days in stock (buy days)	5054	2387	2888	2995	3167	3309
days in T-Bills (sell days)	n.a.	2667	2166	2059	1887	1745
stock return - buy days	.1045	.3492	.2691	.1341	.1506	.1484
- sell days	n.a.	-.1143	-.1147	.0618	.0275	.0216
<u>1970-1974</u>						
days in stock (buy days)	1263	550	733	611	602	604
days in T-Bills (sell days)	n.a.	713	530	652	661	659
stock return - buy days	-.0244	.4606	.2368	.0745	.1079	.0936
- sell days	n.a.	-.3983	-.3853	-.1168	-.1446	-.1323
<u>1975-1979</u>						
days in stock (buy days)	1263	615	699	740	777	844
days in T-Bills (sell days)	n.a.	648	564	523	486	419
stock return - buy days	.1280	.3615	.3125	.1283	.1094	.0620
- sell days	n.a.	-.0933	-.1004	.1280	.1581	.2614
<u>1980-1984</u>						
days in stock (buy days)	1265	570	662	718	744	768
days in T-Bills (sell days)	n.a.	695	603	547	521	497
stock return - buy days	.1300	.3606	.2201	.1378	.1558	.2021
- sell days	n.a.	-.0593	.0310	.1197	.0931	.0184
<u>1985-1989</u>						
days in stock (buy days)	1263	652	794	926	1044	1093
days in T-Bills (sell days)	n.a.	611	469	337	219	170
stock return - buy days	.1845	.2337	.3016	.1753	.2022	.2078
- sell days	n.a.	.1323	-.0132	.2104	.1010	.0359
<u>1990-1995</u>						
days in stock (buy days)	1517	774	820	1000	1202	1230
days in T-Bills (sell days)	n.a.	743	697	517	315	287
stock return - buy days	.1194	.1478	.1337	.0785	.1113	.1032
- sell days	n.a.	.0898	.1027	.1985	.1504	.1887
<u>1988-1992</u>						
days in stock (buy days)	1265	624	680	878	950	980
days in T-Bills (sell days)	n.a.	641	585	387	315	285
stock return - buy days	.1465	.1123	.1330	.0839	.1066	.1224
- sell days	n.a.	.1801	.1626	.2892	.2675	.2303

Table 3
Annual log returns using the subsequent day's closing prices

	Buy and Hold Return	Excess Returns over Buy and Hold				
		AK rule 4	AK rule 7	(1,50,1)	(1,150,1)	(1,200,1)
1970-1989	.1045	-.1187	-.1021	-.0121	.0055	-.0017
<u>Subperiods</u>						
1970-1974	-.0244	-.0321	-.0265	.0789	.1113	.0895
1975-1979	.1280	-.1743	-.1407	-.0449	-.0518	-.0784
1980-1984	.1300	-.1283	-.1534	-.0276	-.0049	.0226
1985-1989	.1845	-.1404	-.0878	-.0552	-.0330	-.0410
1990-1995	.1194	-.1354	-.1251	-.0681	-.0270	-.0336
1988-1992	.1465	-.1669	-.1703	-.0886	-.0606	-.0437

Table 4
1988-1992 Annual log returns using intraday data

	Buy and Hold Return	Excess Returns over Buy and Hold				
		AK rule 4	AK rule 7	(1,50,1)	(1,150,1)	(1,200,1)
Previous closing price*	0.1460	-0.1401	-0.1140	-0.0739	-0.0568	-0.0442
Last trade as of 11:00*	0.1409	-0.1990	-0.1675	-0.0861	-0.0641	-0.0325
Quote midpoint as of 11:00*	0.1409	-0.2014	-0.1693	-0.0873	-0.0640	-0.0328
Current day close*	0.1375	-0.1533	-0.1573	-0.0787	-0.0530	-0.0384
<u>Estimated Execution Prices</u>						
\$10,000,000 initial	0.1407	-0.2176	-0.1860	-0.0895	-0.0653	-0.0336
\$100,000,000 initial	0.1397	-0.3188	-0.2847	-0.1051	-0.0769	-0.0427

* Assumes 0.1% one-way transaction cost.

Table 5
Trading rules using different generating assumptions

Assumptions Used to Generate the 100 Trading Rules					Number Unique	Average (std. dev.)		Across the 100 Trading Rules			
						1970-1989		Pct. of			
index	divi- dends	Trans cost	delay	Excess rule		Return “true”	Trades per yr.	Buy Days	Buy-Sell		
									Return	t-stat.	
<u>Using the AK low-level functions</u>											
1)	S&P	no	0.1%	no	72	.0473 (.010)	-.1014 (.035)	60.9 (19.7)	.626 (.145)	.518 (.107)	6.70 (1.06)
2)	Large	no	0.1%	no	89	.0439 (.016)	-.0956 (.038)	57.2 (21.7)	.641 (.145)	.486 (.150)	6.24 (1.70)
3)	Large	yes	0.1%	no	77	.0391 (.068)	-.0693 (.031)	41.7 (15.5)	.740 (.160)	.535 (.347)	6.27 (4.44)
4)	Large	yes	0.12%	no	66	.0280 (.027)	-.0606 (.025)	37.9 (9.6)	.776 (.129)	.546 (.211)	6.02 (2.01)
5)	Large	yes	0.12%	yes	98	.0009 (.006)	.0009 (.006)	5.0 (1.8)	.650 (.093)	.082 (.073)	1.09 (.41)
<u>With an added function giving current position</u>											
6)	Large	yes	0.12%	yes	85	.0017 (.008)	.0017 (.008)	3.7 (2.7)	.732 (.152)	.086 (.135)	.94 (.72)

Table 6
New trading rules by subperiod

Return measurement period	Average (std. dev.) Across the 100 Trading Rules					
	Number Unique	Excess Return	Trades per yr.	Pct. Buy Days	Buy-Sell	
					Return	t-stat.
1970-1974	67	.0676 (.043)	3.5 (4.8)	.581 (.224)	.138 (.142)	1.03 (.81)
1975-1979	79	-.0537 (.032)	6.0 (3.7)	.751 (.162)	-.289 (.620)	-1.42 (.97)
1980-1984	77	.0132 (.011)	4.1 (2.8)	.725 (.151)	.223 (.598)	.78 (.42)
1985-1989	72	-.0241 (.020)	2.3 (1.7)	.867 (.080)	.152 (.734)	-.04 (.66)
1990-1995	79	-.0386 (.030)	6.8 (5.2)	.805 (.148)	-.118 (.151)	-.90 (.65)
1990-1995*	20	-.0032 (.009)	2.1 (5.9)	.993 (.022)	.005 (.520)	-.04 (.55)

*These rules were generated using 1983-1987 as the training period and 1988-1989 as the selection period.

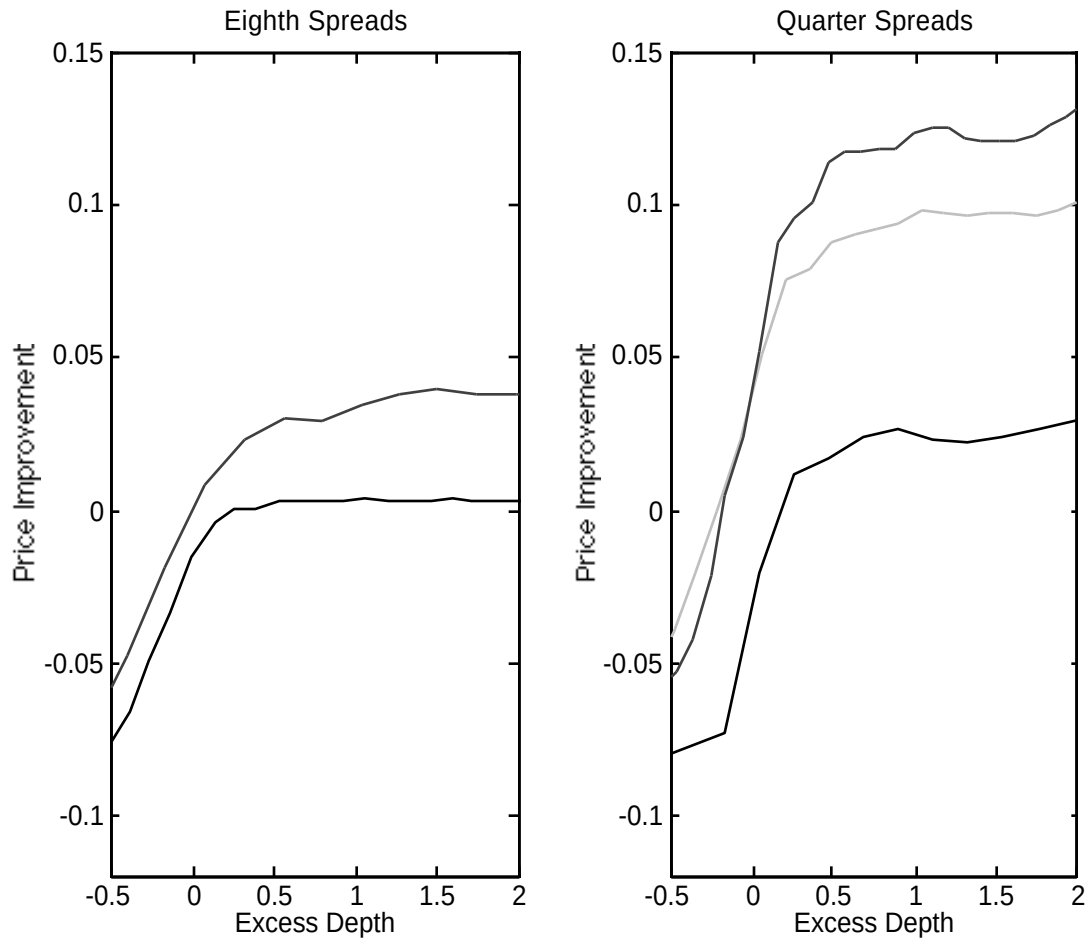


Figure 1 - Loess estimates of expected price improvement.

Price improvement is the difference between execution price and the quoted bid or ask. Excess depth is the difference between the quoted depth and the incoming order, divided by the average quoted depth for the firm. Separate curves are plotted for each possible position of the last trade relative to the current quotes: solid curves reflect “same side”, dashed curves reflect “opposite side,” and the dotted curve reflects “midpoint.”

Endnotes

- ¹ These rules are in the class that BLL call “variable length moving average” rules. They also investigate “fixed length moving average” and “trading range break” rules. The rules that I use are among the most profitable of all of those examined by BLL.
- ² Neely et al (1997) use the AK procedure to study potential profitability of technical trading rules in the foreign exchange markets.
- ³ When the stocks are sold, those with no good quote are assumed to be sold at the previous day’s close, with no transaction cost. In total, 0.7% of the stock sales simulated in Table 4 had missing quote data.
- ⁴ It would be impossible to replicate AK’s results exactly without access to the same random number generator and the initial value of the generator’s “seed.” In addition, there are several minor choices to be made in the implementation of the algorithm, such as the degree to which probabilities are skewed toward good (bad) rules in selecting the rules to be used for (replaced in) the creation of new generations of rules.